

Tech Stack: GPT-4o-mini + LangChain + ChromaDB

HealthLens AI

AI-Powered YouTube Health Video Q&A Assistant



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Problem → Solution → Core Features



Problem

Health videos are long and hard to search; users waste time hunting for specific advice.



Solution

Ask natural-language questions directly about YouTube health videos and get grounded answers.



Core Features

- ✓ Timestamped answers
- ✓ Multi-video library
- ✓ Memory support for follow-ups
- ✓ AI-powered retrieval
- ✓ Speech input/output
- ✓ Multilingual responses



"Instead of watching a 2-hour podcast to find one answer, users can directly ask the AI."

End-to-End System Architecture



Transcript Sources/Tools

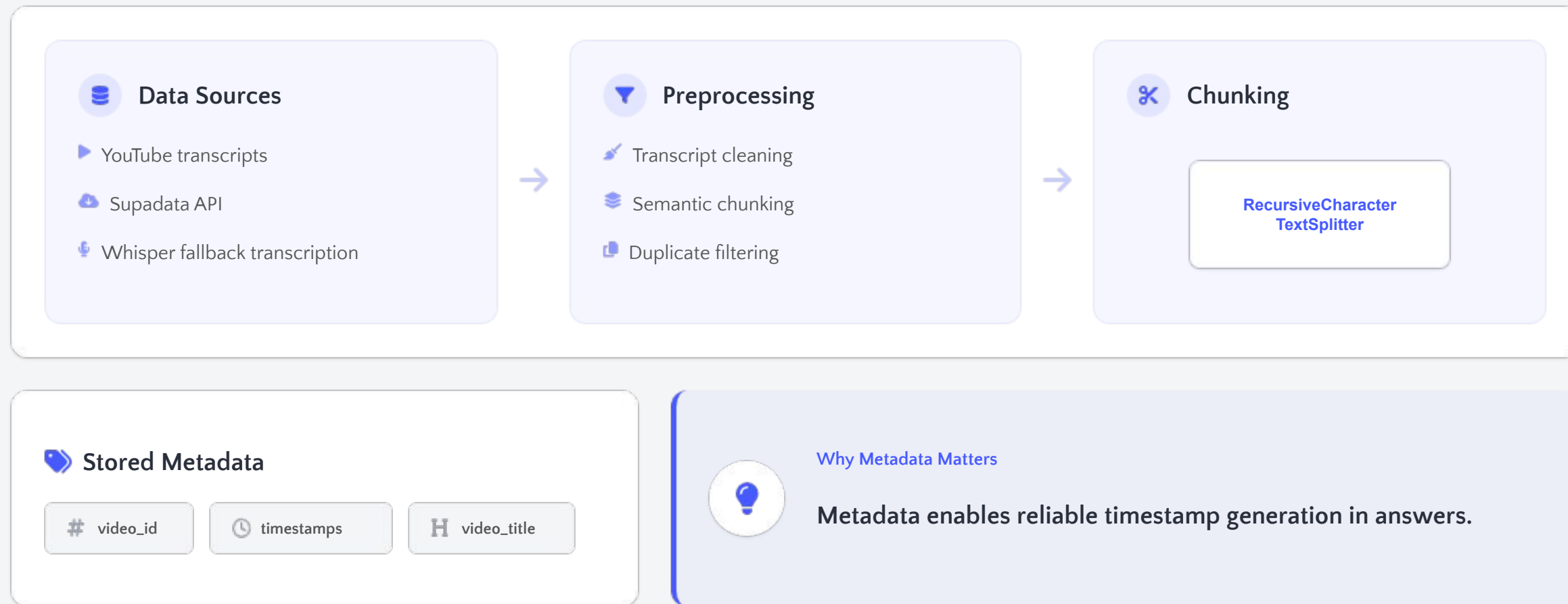
- ✓ **Primary:** Supadata transcript retrieval
- ✓ **Fallback:** Whisper transcription



Output

- 💬 Answer + supporting timestamps

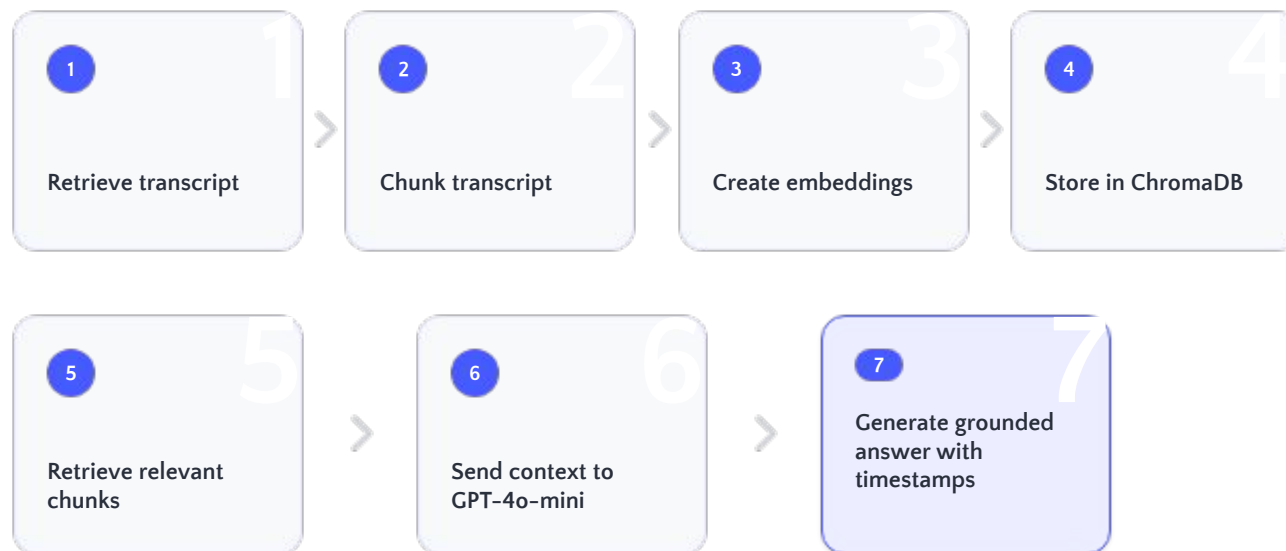
Data Sources, Chunking, and Metadata



Retrieval-Augmented Generation (RAG) Pipeline



7-Step Flow



Guardrails

- ✓ `similarity_search_with_score()`
- ✓ Similarity threshold filtering
- ✓ Duplicate chunk filtering
- ✓ Timestamp source generation
- ✓ Hallucination-protection prompts

LangChain Agent: Routing, Memory, Language Matching

Agent System

- Router prompt
- Multi-tool architecture
- PromptTemplates
- Conversation memory
- Language matching



Memory

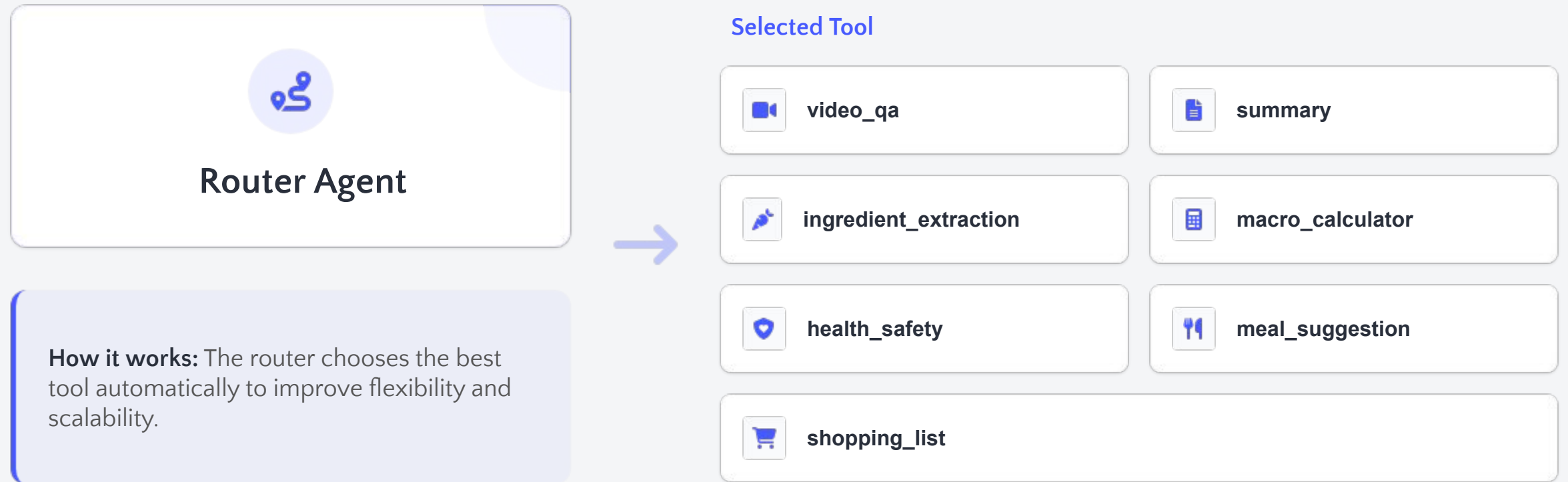
Stores recent conversation history to support follow-up questions.



Language Support

If the user asks in another language, the AI answers in the same language automatically.

Implemented Tooling (Router-Selected)



Speech Input/Output for Accessibility

Speech Features

- ✓ Speech-to-text (user asks by voice)
- ✓ Text-to-speech (AI reads answers)
- ✓ Browser speech synthesis
- ✓ Stop-reading functionality

Why it matters

- ✓ Better usability on mobile
- ✓ Accessibility support



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Evaluation Setup (LLM-as-a-Judge)

Evaluation Method

GPT-4o-mini used as LLM-as-a-judge

Tools Used

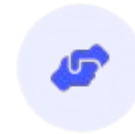
- ✓ LangSmith tracing
- ✓ Terminal evaluation pipeline



Grounding Note:

Evaluations were grounded on retrieved sources.

Metrics



Correctness



Answer Relevance



Bias & Fairness



Hallucination

Results and Key Weaknesses



Single-Video RAG

- Most answers scored 5/5 across metrics
- Strong hallucination safety



Library Mode

- Works well for specific questions
- Harder for broad cross-video retrieval

Main Weakness: Chronological reasoning—semantic retrieval struggles with "what came first" questions.

"Semantic retrieval does not always understand temporal ordering."

LangSmith Traces (What We Inspected)

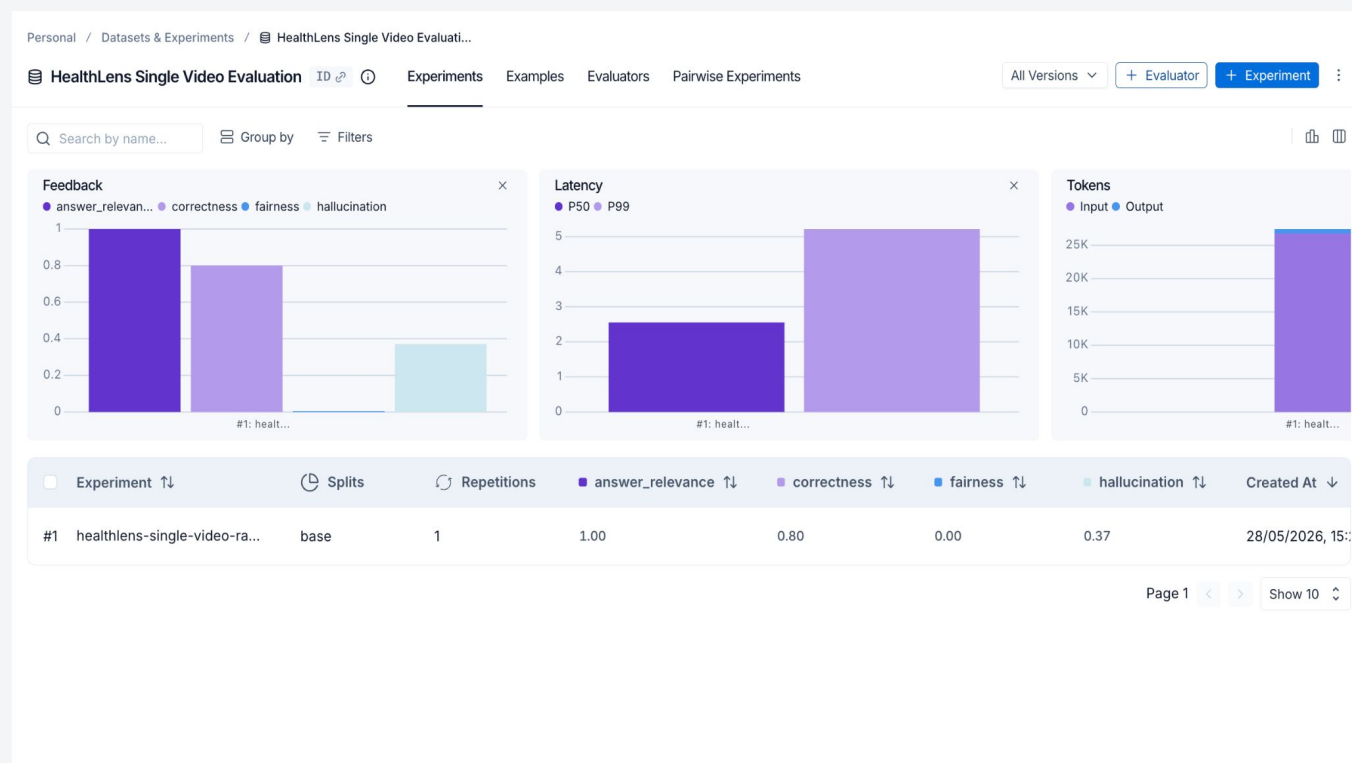
Trace View Focus

- ✓ User question
- ✓ Retrieved chunks
- ✓ Prompt
- ✓ Final answer
- ✓ Tool execution
- ✓ Token usage
- ✓ Latency



Why it helped:

LangSmith made it easier to debug and evaluate the agent end-to-end.



Terminal Tests

Question	Answer	Relevance	Correctness	 Bias & Fairness	Hallucination
What is the main topic of the video?	The video is about six habits that wreck sleep and how to improve sleep quality.	5/5	5/5	5/5	5/5
Which habit was mentioned first?	The first habit mentioned is eating food too late.	5/5	1/5	5/5	1/5
What does the video say about caffeine?	Caffeine late in the day disrupts sleep because its 6-hour half-life leaves stimulation active at bedtime.	5/5	5/5	5/5	5/5
What does the video say about bedroom temperature?	A bedroom that is too hot or too cold disrupts sleep, and the speaker recommends cooling strategies and temperature-controlled sleep systems.	5/5	5/5	5/5	5/5
What does the video say about noise and sleep?	Noise like barking dogs or arguing neighbors can interrupt sleep, and white/brown/pink noise machines may help reduce disruptions.	5/5	5/5	5/5	5/5
What does the video say about eating too late?	Eating too late causes digestion activity, glucose fluctuations, and melatonin disruption that negatively affect sleep.	5/5	5/5	5/5	5/5
What does the video recommend for winding down before bed?	Reading, journaling, meditating, baths, and walks are recommended as calming nighttime activities.	5/5	5/5	5/5	5/5
What does the video say about light exposure before sleep?	The video recommends dimming lights, reducing blue light, and avoiding screens before bedtime.	5/5	5/5	5/5	5/5
What are the top 3 sleep mistakes mentioned?	Eating too late, not having a wind-down routine, and using stimulants too late in the day.	5/5	5/5	5/5	5/5
What does the video say about drinking alcohol before sleep?	The speaker stopped alcohol because it disrupted mood stability and negatively affected sleep.	5/5	5/5	5/5	5/5

Deployment and Security



Deployment Stack

- ✓ Frontend: HTML/CSS/JavaScript
- ✓ Backend: FastAPI
- ✓ Hosting: Render
- ✓ Source: GitHub repository



Security Practices

- ✓ .env protection
- ✓ .gitignore usage
- ✓ API keys hidden



Local-first development with deployment-ready architecture.

Future Improvements



Better chronological reasoning



Hybrid search



Better cross-video retrieval



Reranking system



Larger video library



Advanced memory

Thank You

 [Live Demo](#)

Demo Questions:

- 1 "What does the video say about caffeine?"
 - 2 "What does the library say about McDonald's food?"
 - 3 "What does the video say about light exposure before sleep?"
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